



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE (Advanced)

FULL SYLLABUS

27-09-2021

JEE(Main + Advanced) : LEADER & ENTHUSIAST COURSE - SCORE(ADVANCED)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
A.		0.78 to 0.79	2.40	55.00	1.38 to 1.40	0.26 to 0.27	200.00	3.33	0.50	500.00	4.57
SECTION-III	Q.	1	2	3	4	5	6	7	8	9	10
A.		3	8	1	2	1	2	1	2	6	4

PART-2 : CHEMISTRY

SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
A.		11.75	0.78	1.58	0.04	1.60	2.00	6.00	7.00	5.00	2.00
SECTION-III	Q.	1	2	3	4	5	6	7	8	9	10
A.		3	2	6	3	5	2	5	5	1	1

PART-3 : MATHEMATICS

SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
A.		0.50	0.50	1.25	1.33 or 1.34	3.46	0.50	0.80	0.23	0.16 or 0.17	6.50
SECTION-III	Q.	1	2	3	4	5	6	7	8	9	10
A.		5	2	3	1	2	4	2	3	1	8

HINT – SHEET

PART-1 : PHYSICS

SECTION-II

1. Ans (0.78 to 0.79)

Image for lens is 10 cm behind mirror

$$u = 10 \cos \theta \approx 10 \quad h_0 = 10 \sin \theta$$

$$\frac{1}{v} + \frac{1}{10} = \frac{1}{50}$$

$$\frac{1}{v} = \frac{1}{50} - \frac{1}{10} = \frac{1-5}{50}$$

$$v = -12.5 \text{ cm}$$

$$h_1 = h_0 \times -\frac{v}{u} = 10 \sin \theta \times \frac{12.5}{10} = 12.5 \theta$$

$$y = 12.5 \theta + 12.50 \cos \theta = \frac{25\pi}{100} = \frac{3.14}{4} \text{ cm}$$

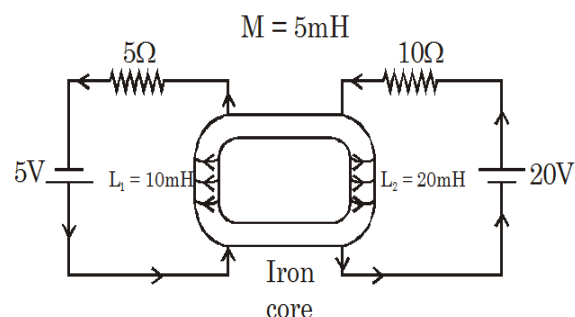
2. Ans (2.40)

$$R = \frac{\rho}{2\pi R_1 R_2} (R_2 - R_1)$$

$$\frac{4\pi \times 0.1}{2\pi \times 0.1 \times 0.2} = 10\Omega$$

$$i = 2.4 \text{ A}$$

3. Ans (55.00)



B due to both is in same direction

$$W = \frac{1}{2} L_1 i_1^2 + \frac{1}{2} L_2 i_2^2 + M i_1 i_2$$

4. Ans (1.38 to 1.40)

$$i_1 = \frac{V}{R} \left(1 - e^{-\frac{Rt}{L}} \right)$$

$$i_2 = \frac{V}{R} e^{-t/RC}$$

$$i = i_1 - i_2 = 0$$

$$i_1 = i_2$$

$$\Rightarrow 1 - e^{-t/T} = e^{-t/T}$$

$$1 = 2e^{-t/T}$$

$$\ln 2 = \frac{t}{\tau} \Rightarrow t = 2 \ln 2$$

5. **Ans (0.26 to 0.27)**

$$a_r = \frac{g \sin \theta}{1 + \frac{1}{mR^2}} = \frac{5}{2} \text{ m/s}^2$$

$$a_d = \frac{5}{3/2} = \frac{10}{3} \text{ m/s}^2$$

$$5 = \frac{1}{2} \times \frac{5}{2} \times t_r^2$$

$$5 = \frac{1}{2} \times \frac{10}{3} \times t_d^2$$

$$\Rightarrow t_r = 2 \text{ sec}$$

$$t_d = \sqrt{3} \text{ sec.}$$

$$\Delta t = 2 - \sqrt{3} = 0.268 \approx 0.26 \text{ to } 0.27$$

6. **Ans (200.00)**

$$\frac{4k}{L} \times \pi r^2 (300 - T) = \frac{k}{L} \pi (2r)^2 (T - 100)$$

$$\Rightarrow T = 200$$

7. **Ans (3.33)**

$$C_2 = \frac{\frac{\epsilon_0 S}{2}}{\frac{d}{2 \times 2} + \frac{d}{2 \times 4}} + \frac{\frac{4\epsilon_0 S}{2}}{d}$$

$$= \frac{\epsilon_0 S}{d} \left[\frac{4}{3} + 2 \right] \Rightarrow \frac{C_2}{C_1} = \frac{10}{3} = 3.33$$

8. **Ans (0.50)**

$$2T \sin \frac{\theta}{2} = iR\theta B$$

$$\Rightarrow T = iRB = \frac{iBL}{2\pi} = \frac{2 \times 1.57 \times 1}{2 \times 3.14} = 0.5 \text{ N}$$

9. **Ans (500.00)**

$$P_0 V_0 = PV$$

$$10^5 \times 0.475 \times A = (10^5 - 10^3 \times 10 \times y) (1 - y)A$$

$$4.75 = (10 - y) (1 - y) = 10 + y^2 - 11y$$

$$y^2 - 11y + 5.25 = 0$$

$$y = \frac{11 \pm \sqrt{121 - 21}}{2} = 10.5 \text{ or } 0.5 \text{ m}$$

10. **Ans (4.57)**

$$Q = (M_{Ra} - M_{Rm} - M_{\alpha}) C^2$$

$$= \frac{931 \times 5}{1000} = 4.655 \text{ MeV}$$

$$k_{\alpha} = \frac{A - 4}{A} Q = \frac{222}{226} \times 4.655 = 4.57 \text{ MeV}$$

SECTION-III

1. **Ans (3)**

$$B = \frac{\mu_0 J x}{2}$$

$$B_{\text{due to cylinder}} = 0 \text{ as } x = 0$$

$$B_{\text{cavity}} = \frac{\mu_0 J \frac{a}{2}}{2} = \frac{\mu_0 J a}{4} \Rightarrow N = 3$$

2. **Ans (8)**

$$T \cos \theta = mg$$

$$T \sin \theta = m\omega^2 r = m\omega^2 L \sin \theta$$

$$\omega = \sqrt{\frac{T}{mL}} = 8 \text{ rad/s}$$

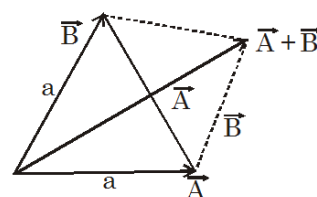
3. **Ans (1)**

It forms wheatstone bridge

$$R_{eq} = 3.5 + 4.5 = 8\Omega$$

$$i = \frac{8}{8} = 1 \text{ A}$$

4. **Ans (2)**



$$|\vec{A} + \vec{B}| = |\vec{A} - \vec{B}|$$

$$\text{when } \omega t = \frac{\pi}{2}$$

$$\frac{\pi}{4} \tau = \frac{\pi}{2}$$

$$\Rightarrow t = 2 \text{ sec.}$$

5. **Ans (1)**

$$\Delta x = d \sin \alpha + \frac{dy}{D}$$

$$d\alpha = \Delta x - \frac{dy}{D} \Rightarrow \alpha = \frac{\Delta x}{d} - \frac{y}{D}$$

$$\alpha = \frac{n \times 6 \times 10^{-7}}{3 \times 10^{-4}} - \frac{11 \times 10^{-3}}{1}$$

$$= 2n \times 10^{-3} - 11 \times 10^{-3} = 10^{-3} [2n - 11]$$

$$2n - 11 > 0$$

$$n > 5.5 \times 10^{-3}$$

$$n = 6 = 10^{-3} \text{ rad.}$$

6. **Ans (2)**

$$v_{\text{esc}} = \sqrt{2gR} = 11$$

$$v'_{\text{esc}} = \sqrt{\frac{2g}{11}} \times R'$$

$$g = \frac{4}{3\pi} G \rho R$$

$$\frac{1}{11} \times \frac{4}{3} \pi G \rho R = \frac{g}{11} = \frac{4}{3} \pi G \frac{\rho}{4} R'$$

$$R' = \frac{4}{11} R$$

$$\Rightarrow v'_{\text{sec}} = \sqrt{\frac{2g}{11} \times \frac{4R}{11}} = \frac{2}{11} \times 11 = 2 \text{ km/s}$$

7. **Ans (1)**

$$N = N_0 e^{-t/T}$$

$$N_d = N_0 \left(1 - e^{-\frac{t}{T}}\right)$$

$$\ln \left(1 - \frac{N_d}{N_0}\right) = -\frac{t}{T}$$

$$T = \frac{t}{\ln \left(\frac{N_0}{N_0 - N_d}\right)}$$

$$dT = \frac{t}{\left(\ln \left(\frac{N_0}{N_0 - N_d}\right)\right)^2} \times \left(0 + \frac{dN_d}{N_0 - N_d}\right)$$

$$dT = \frac{49}{\ln(2)^2} \times \frac{10}{1000} = 1 \text{ sec.}$$

8. **Ans (2)**

$$6.28 = \frac{2\pi}{\lambda} = 62.8$$

$$\lambda = 0.1 \text{ m}$$

$$\ell = 40 \frac{\lambda}{2} = 2 \text{ m}$$

9. **Ans (6)**

$$F = \frac{dN}{dt} \times mv$$

$$= \frac{\rho \lambda}{hc} \times \sqrt{2mk}$$

$$= \frac{\rho \lambda}{hc} \times \sqrt{2mqv}$$

$$\frac{6.25 \times 1.6 \times 10^{-19}}{\sqrt{2 \times 9 \times 10^{-31} \times 1.6 \times 10^{-19} \times 125}}$$

$$1 \times 10^{20} \times 10^{-25} \times \sqrt{25 \times 9 \times 16}$$

$$= 60 \times 10^{-5} = 6 \times 10^{-4} \text{ N}$$

10. **Ans (4)**

$$a = -\mu g = -2 \text{ m/s}^2$$

$$1.6 = 0.4 v$$

$$v = 4 \text{ m/s}$$

comes to rest at

$$4 - 2t = 0$$

$$t = 2 \text{ sec.}$$

$$\text{so } s = \frac{u^2}{2a} = \frac{16}{2 \times 2} = 4 \text{ m}$$

PART-2 : CHEMISTRY

SECTION-II

1. **Ans (11.75)**

$$E_{ab} - E_{af} = 4RT ; A_f = 2 \times A_b$$

$$K_f = A_f e^{\frac{-E_{af}}{RT}} \dots\dots\dots(1)$$

$$K_b = A_b e^{\frac{-E_{ab}}{RT}} \dots\dots\dots(2)$$

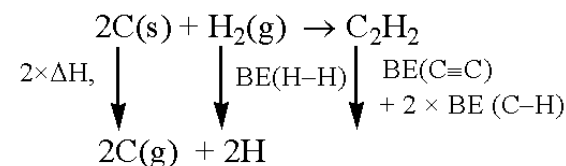
$$K_{eq} = \frac{K_f}{K_b} = \frac{A_f}{A_b} e$$

$$K_{eq} = 2e^4$$

$$\Delta^\circ G = -RT \ln K_{eq} = -2500 \times [4.7]$$

$$= 11.75 \text{ kJ/mol}$$

2. **Ans (0.78)**



$$\Delta H_{(rxn)} = \Delta_f H(C_2H_2) = [2 \times \Delta H_f + \text{BE(H-H)}] - [\text{BE(C}\equiv\text{C)} + 2 \times \text{BE(C-H)}]$$

$$\Rightarrow 220 = (1400 + 300) - (\text{BE(C}\equiv\text{C)} + 2 \times 350)$$

$$\Rightarrow \text{BE(C}\equiv\text{C)} = 780 \text{ kJ/mol.}$$

3. **Ans (1.58)**

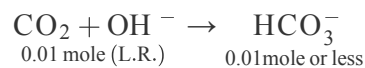
$$Q_{\text{cell}} = \frac{[Fe^{+2}]^2}{P_{O_2} \times [H^+]^4} = \frac{(10^{-3})^2}{(0.01)(10^{-4})^4} = 10^{12}$$

$$E = E^0 - \frac{0.06}{4} \log 10^{12} = 1.76 - 0.18 = 1.58 \text{ V}$$

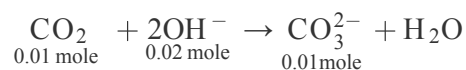
4. Ans (0.04)

$$n_{\text{CO}_2} = \frac{224}{22400} = 0.01$$

$$\text{and } n_{\text{H}^+ \text{ used}} = \frac{30 \times 1}{1000} = 0.03$$



Hence, moles of H^+ used should be 0.01 or less and titration of HCO_3^- and H^+ should not be detected by phenolphthalein. Hence, OH^- must be in excess.



Thus, 0.01 mole of CO_3^{2-} will require only 0.01 mole of H^+ in the presence of phenolphthalein. As the mole of H^+ used is 0.03, 0.02 mole OH^- must be present in excess.

Hence, total moles of OH^- used

$$= 0.02 + 0.02 = 0.04,$$

$$\therefore [\text{NaOH}]_{\text{used}} = \frac{0.04}{1} = 0.04\text{M}$$

5. Ans (1.60)

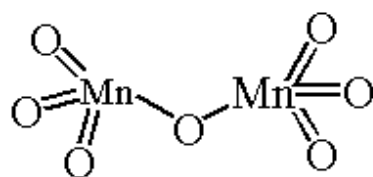
$$x = 5, y = 3 (\text{O}_2, \text{CCl}_4, \text{C}_6\text{H}_6)$$

6. Ans (2.00)



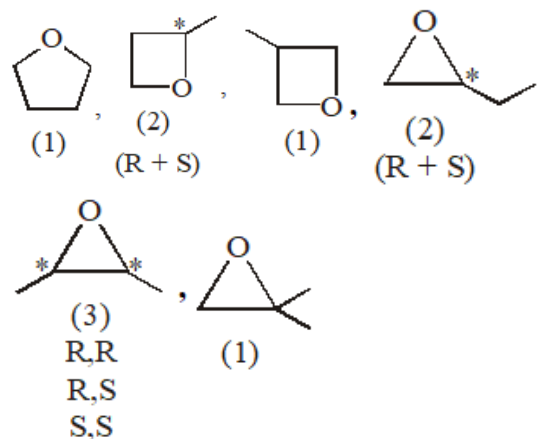
$\searrow$
 sp^3d^2 hybridised

7. Ans (6.00)

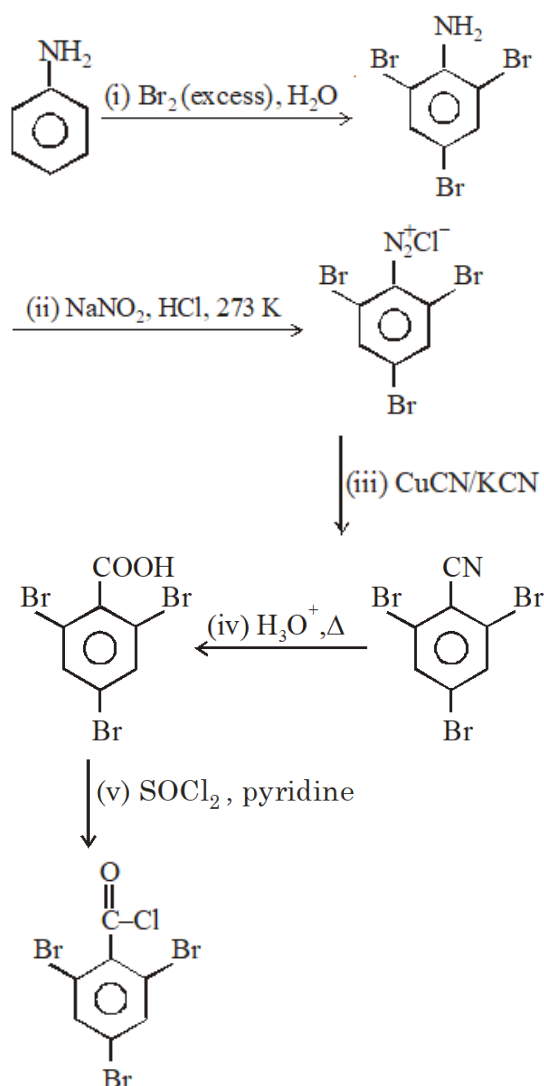


All Mn = 0

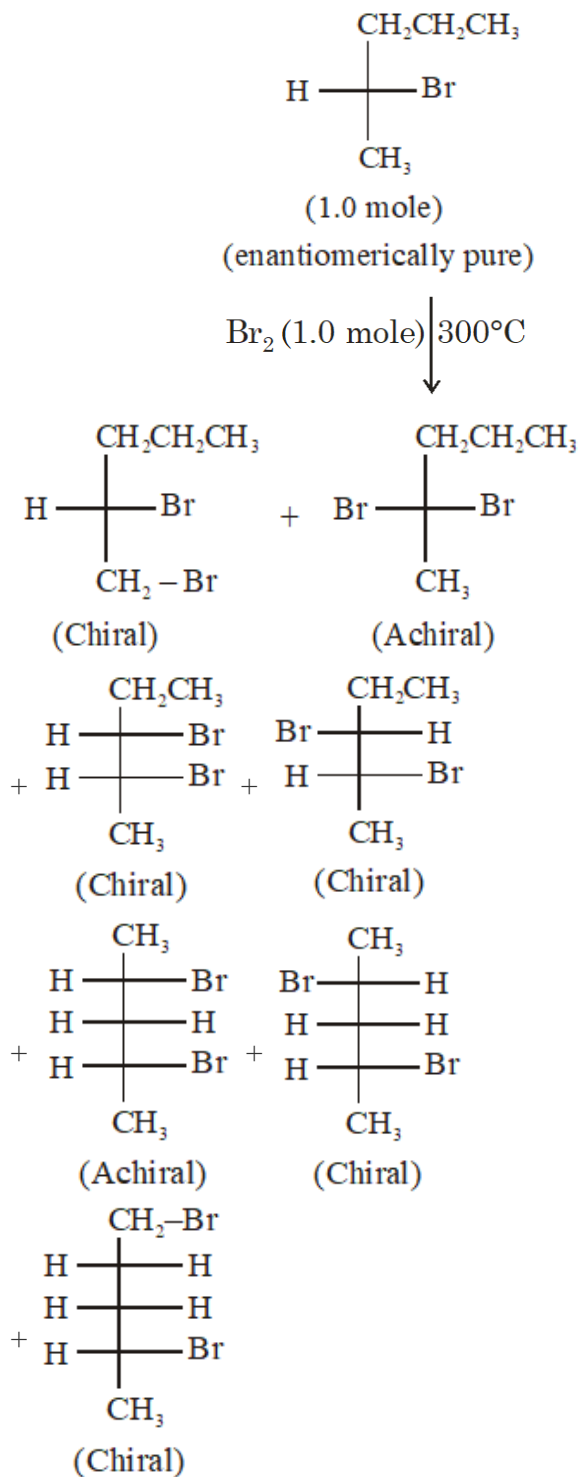
8. Ans (7.00)



9. Ans (5.00)



10. Ans (2.00)



SECTION-III

1. Ans (3)

Pressure exerted by He : $1 - 0.68 = 0.32 \text{ atm}$
 $PV = nRT$
 $\Rightarrow (0.32 \text{ atm}) V = 0.04 \text{ mol} \times 0.08 \times 300 \text{ K}$
 $\Rightarrow V = 3 \text{ L}$

2. Ans (2)

$$\begin{aligned}
 \text{Density} &= 8 = \frac{Z_{\text{eff}} \times M}{a^3 \times N_A} \\
 \Rightarrow M &= \frac{8 \times (200 \times 10^{-10})^3 \times 6 \times 10^{23}}{4} \\
 &= 9.6 \text{ g/mol} \\
 \therefore \text{No. of atoms in 32 gm} &= \left(\frac{32}{76.8} \times 6 \times 10^{23} \right) \\
 &= 2 \times 10^{24} \\
 N &= 2
 \end{aligned}$$

3. Ans (6)

$$\text{Molarity} = 2.4 = \frac{n_{\text{soluble}}}{V_{\text{solution}} \text{ (in } \ell \text{)}}$$

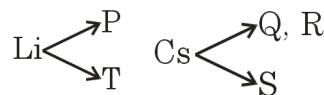
Assuming 1 l ; solution = 400 g

$\Rightarrow$ mass of solvent = 400 g

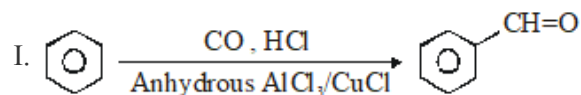
$\Rightarrow n_{\text{solute}} = 2.4$

$$\Rightarrow \text{molality} = \frac{2.4}{0.4} = 6$$

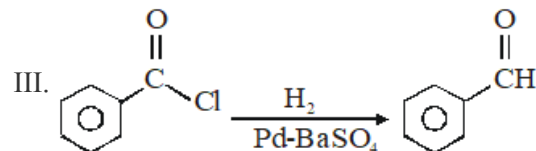
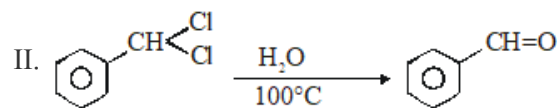
6. Ans (2)



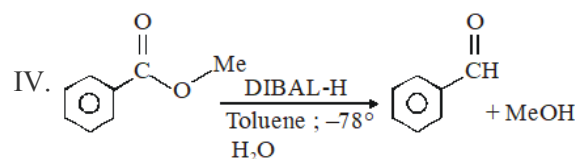
7. Ans (5)



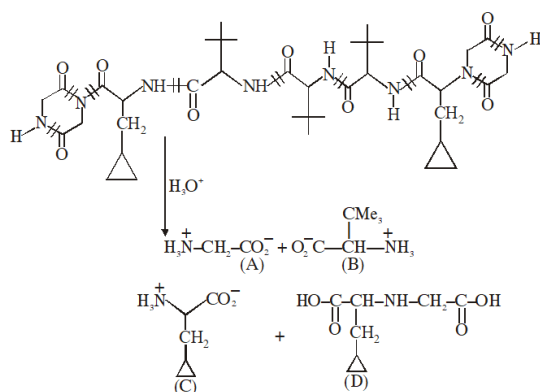
this reaction is called Gattermann koch synthesis



this reaction is called Rosenmund reduction

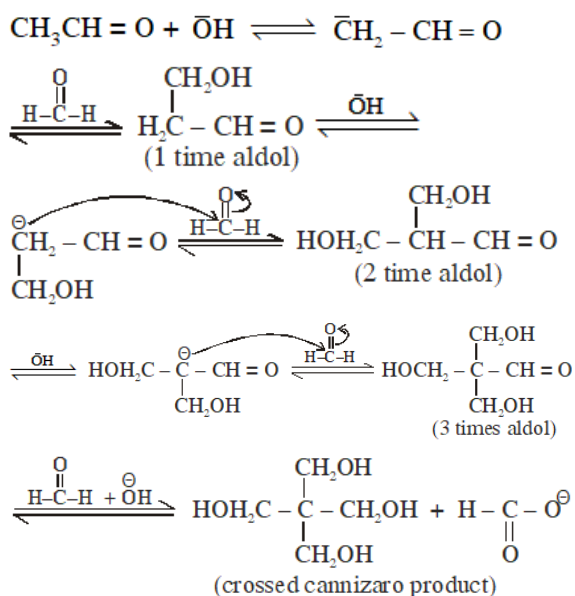


9. Ans (1)



A $\Rightarrow$ is glycine which is only naturally occurring amino acid.

10. Ans (1)



PART-3 : MATHEMATICS

SECTION-II

1. Ans (0.50)

$$\text{Now, } \sum_{r=0}^{2n} a_r (x-2)^r = \sum_{r=0}^{2n} b_r (x-3)^r$$

On differentiating both sides n times w.r.t. x ,

we get

$$\sum_{r=n}^{2n} a_r (n!) {}^r C_n (x-2)^{r-n} = \sum_{r=n}^{2n} b_r (n!) {}^r C_n (x-3)^{r-n}$$

On putting $x = 3$,

$$\text{we get } \sum_{r=n}^{2n} a_r (n!) {}^r C_n = (b_n) n!$$

[since, all the terms except first on RHS become zero]

$$\Rightarrow b_n = {}^n C_n + {}^{n+1} C_n + {}^{n+2} C_n + \dots + {}^{2n} C_n$$

$$[\because a_r = 1, \forall r \geq n]$$

$$= {}^{2n} C_{n+1} + {}^{2n} C_n = {}^{2n+1} C_{n+1}$$

2. Ans (0.50)

Let 7 number be $x_1, x_2, x_3, \dots, x_7$ and these number

can be chosen from 1 to 100. and $|x_i - x_j| \geq 7$

$$\boxed{}_{d_8}^{x_1} \boxed{}_{d_1}^{x_2} \boxed{}_{d_2}^{x_3} \boxed{}_{d_3}^{x_4} \boxed{}_{d_4}^{x_5} \boxed{}_{d_5}^{x_6} \boxed{}_{d_6}^{x_7} \boxed{}_{d_7}$$

gap between any two number be $d_1 d_2 \dots d_6 d_7$

$$d_1 + d_2 + d_3 + d_4 + d_5 + d_6 + d_7 + d_8 = 93$$

where $d_1 \geq 7, d_2 \geq 7, d_3 \geq 7, d_4 \geq 7, d_5 \geq 7, d_6 \geq 7, d_7 \geq 7,$

$$d_8 \geq 7$$

$$\text{so } d_i = t_i + 7, t_i \geq 0 \ i \in \{1, 2\}$$

$$t_1 + t_2 + t_3 + t_4 + t_5 + t_6 + t_7 + t_8 = 93 - 7 \times 6 = 51$$

number of non negative integral solution

$$= {}^{51+8-1} C_7 = {}^{58} C_7$$

$$n = 58 \Rightarrow n = 2^1 \times 29^1$$

3. **Ans (1.25)**

Let m and $(m + 1)$ be the removed numbers from $1, 2, \dots, n$. Then, the sum of the remaining numbers is

$n(n + 1)/2 - (2m + 1)$. From given condition,

$$\frac{105}{4} = \frac{\frac{n(n+1)}{2} - (2m + 1)}{(n - 2)}$$

$$\text{or } 2n^2 - 103n - 8m + 206 = 0$$

Since n and m are integers, so n must be even.

Let $n = 2k$. Then,

$$m = \frac{4k^2 + 103(1 - k)}{4}$$

Since m is an integer, then $1 - k$ must be divisible by 4. Let $k = 1 + 4t$.

Then we get $n = 8t + 2$,

$$m = 16t^2 - 95t + 1. \text{ Now,}$$

$$1 \leq m < n$$

$$\Rightarrow 1 \leq 16t^2 - 95t + 1 < 8t + 2$$

Solving, we get $t = 6$. Hence,

$$n = 50 \text{ and } m = 7$$

4. **Ans (1.33 or 1.34)**

$$\text{Given } \frac{xdy}{dx} - 2y = x^4y^2 \text{ or } \frac{dy}{dx} - \frac{2}{x}y = x^3y^2$$

. Now dividing by y^2 , we get

$$\frac{1}{y^2} \frac{dy}{dx} - \frac{2}{y} \frac{1}{x} = x^3$$

substituting $\frac{-2}{y} = t$, we get

$$\Rightarrow \frac{1}{2} \frac{dt}{dx} + \frac{t}{x} = x^3 \Rightarrow \frac{dt}{dx} + \frac{2}{x}t = 2x^3$$

$$\text{I.F.} = 2 \int \frac{2}{x} dx = e^{2\ln x} = e^{\ln x^2} = x^2$$

So, general solution is given by

$$x^2 t = \frac{2x^6}{6} + C \Rightarrow \frac{-2}{y} = \frac{x^4}{3} + \frac{C}{x^2}$$

$$\text{If } x = 1, y = -6, \Rightarrow C = 0$$

$$\frac{-2}{y} = \frac{x^4}{3} \Rightarrow y = \frac{-6}{x^4} \text{ i.e., } f(x) = \frac{-6}{x^4}$$

$$\text{Now, } \frac{dy}{dx} = 24x^{-5} = \frac{24}{x^5}. \text{ Hence}$$

$$\left[\frac{dy}{dx} \right]_{x=3^{1/5}} = \frac{24}{3} = 8$$

5. **Ans (3.46)**

$A(z_1)$ is the point of intersection of $\arg z$

$$(z - 2 + i) = \frac{3\pi}{4} \text{ and}$$

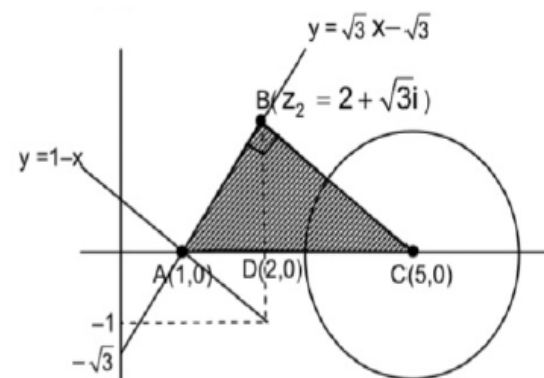
$$\arg(z + \sqrt{3}i) = \frac{\pi}{3} \Rightarrow z_1 = 1$$

$$B(z_2) \text{ is the point on } \arg(z + \sqrt{3}i) = \frac{\pi}{3}$$

Such that $|z_2 - 5|$ is minimum Hence

$$z_2 = 2 + \sqrt{3}i$$

$C(z_3)$ be the centre of the circle



$$|z - 5| = 3 \Rightarrow z_3 = 5$$

$$\Rightarrow \text{Area of } \triangle ABC = \frac{1}{2} \times AC \times BD = \frac{1}{2} \cdot 4 \cdot \sqrt{3}$$

$$\sqrt{3} = 2\sqrt{3} \text{ sq. units.}$$

6. **Ans (0.50)**

$$\text{Given, } \cos A \cos B + \sin A \sin B \sin C = 1$$

$$\Rightarrow \sin C = \frac{1 - \cos A \cos B}{\sin A \sin B} \dots(i)$$

$$\Rightarrow \frac{1 - \cos A \cos B}{\sin A \sin B} \leq 1 \quad [\because \sin C \leq 1]$$

$$\Rightarrow 1 \leq \cos(A - B)$$

$$\Rightarrow \cos(A - B) \geq 1$$

$$\Rightarrow \cos(A - B) = 1 \quad [\because \text{as } \cos(\theta) \leq 1]$$

$$\Rightarrow A - B = 0$$

On putting $A = B$ in Eq.(i), we get

$$\sin C = \frac{1 - \cos^2 A}{\sin^2 A}$$

$$\Rightarrow \sin C = 1$$

$$\Rightarrow C = \frac{\pi}{2}$$

$$\text{Now, } A + B + C = \pi$$

$$\Rightarrow \sin A : \sin B : \sin C = \sin \frac{\pi}{4} : \sin \frac{\pi}{4} : \sin \frac{\pi}{2}$$

$$\Rightarrow a : b : c = \frac{1}{\sqrt{2}} : \frac{1}{\sqrt{2}} : 1$$

$$= 1 : 1 : \sqrt{2}$$

7. **Ans (0.80)**

$$\text{Let } I = \int_0^1 \log(\sqrt{1-x} + \sqrt{1+x}) dx$$

$$\text{Put } x = \cos 2\theta$$

$$\Rightarrow dx = -2\sin\theta d\theta$$

$$= -2 \int_{\pi/4}^0 \log[\sqrt{2}(\sin\theta + \cos\theta)] \sin 2\theta d\theta$$

$$= -2 \int_{\pi/4}^0 [(\log \sqrt{2}) \sin 2\theta + \log(\sin\theta + \cos\theta) \cdot \sin 2\theta] d\theta$$

$$= -2 \log \sqrt{2} \left[\frac{-\cos 2\theta}{2} \right]_{\pi/4}^0$$

$$-2 \int_{\pi/4}^0 \log(\sin\theta + \cos\theta) \cdot \sin 2\theta d\theta$$

$$= \log \sqrt{2} - 2 \left[- \left\{ \log(\sin\theta + \cos\theta) \cdot \frac{\cos 2\theta}{2} \right\} - \int_{\pi/4}^0 \left(\frac{\cos\theta - \sin\theta}{\cos\theta + \sin\theta} \times \frac{-\cos 2\theta}{2} \right) d\theta \right]$$

$$= \log(\sqrt{2}) - 2 \left[0 + \frac{1}{2} \int_{\pi/4}^0 (\cos\theta - \sin\theta)^2 d\theta \right]$$

$$= \frac{1}{2} \log 2 - \left[\theta + \frac{\cos 2\theta}{2} \right]_{\pi/4}^0$$

$$= \frac{1}{2} \log 2 - \left(\frac{1}{2} - \frac{\pi}{4} \right) = \frac{1}{2} \log 2 - \frac{1}{2} + \frac{\pi}{4}$$

8. **Ans (0.23)**

Hence, the total number of favourable cases

$$= (3^n - 3 \times 2^n + 3) \times {}^6C_3$$

$$\therefore \text{Required probability} = \frac{(3^n - 3 \cdot 2^n + 3) \times {}^6C_3}{6^n},$$

Here $n = 6$

9. **Ans (0.16 or 0.17)**

Let the equation of plane through (1,1, 1)

having a, b, c

as DR's of normal to plane,

$$a(x-1) + b(y-1) + c(z-1) = 0$$

and plane is parallel to straight line having

DR's.

$$(1,0,-1) \text{ and } (-1,1,2)$$

$$\Rightarrow a - c = 0$$

$$\text{and } -a + b + 2c = 0$$

$$\Rightarrow b = -c$$

$\therefore$ Equation of plane is

$$c(x-1) - c(y-1) + c(z-1) = 0$$

$$x-1-y+1+z-1=0$$

$$x-y+z=1$$

Its intercept on coordinate axes are

$$A(1,0,0), B(0,-1,0), C(0,0,1)$$

Hence, the volume of tetrahedron OABC

$$= \frac{1}{6} [abc] = \frac{1}{6} \begin{vmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = \frac{1}{6} = \frac{1}{6} \text{ cu units}$$

10. **Ans (6.50)**

$$\int (x^{2010} + x^{804} + x^{402}) (2x^{1608} + 5x^{402} + 10)^{\frac{1}{402}} dx$$

By rearranging terms

$$\int (x^{2009} + x^{803} + x^{401}) (2x^{2010} + 5x^{804} + 10x^{402})^{\frac{1}{402}} dx$$

$$\text{Put } 2x^{2010} + 5x^{804} + 10x^{402} = t$$

$$\Rightarrow (4020x^{2009} + 4020x^{803} + 4020x^{401}) dx$$

$$= dt \Rightarrow (x^{2009} + x^{803} + x^{401}) dx = \frac{dt}{4020}$$

$$\int \frac{dt}{4020} t^{\frac{1}{402}} \Rightarrow \frac{t^{\frac{403}{402}}}{4020 \times \frac{403}{402}} + c = \frac{t^{\frac{403}{402}}}{4030} + c$$

hence $a = 403$

SECTION-III

1. Ans (5)

$$|\vec{a} + \vec{b} + \vec{c}|^2 \geq 0$$

$$\therefore \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} \geq \frac{-1}{2} (|\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2)$$

$$\text{Given, } |\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2 = 9$$

$$\Rightarrow |\vec{a}|^2 + |\vec{b}|^2 - 2\vec{a} \cdot \vec{b} + |\vec{b}|^2 + |\vec{c}|^2 - 2\vec{b} \cdot \vec{c} + |\vec{c}|^2 + |\vec{a}|^2 - 2\vec{c} \cdot \vec{a} = 9$$

$$\Rightarrow \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = -\frac{3}{2} \quad \dots(i)$$

Also,

$$\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} \geq \frac{-1}{2} (|\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2) \geq -\frac{3}{2} \quad \dots(ii)$$

$$\text{From Eqs. (i) and (ii), } |\vec{a} + \vec{b} + \vec{c}| = 0$$

$$\Rightarrow \vec{a} + \vec{b} + \vec{c} = \vec{0}$$

$$|2\vec{a} + 5\vec{b} + 5\vec{c}| = |2\vec{a} - 5\vec{a}| = 3$$

2. Ans (2)

Since the line touches the graph of $f(x)$ at 1,2,3

$$\therefore h(x) = f(x) - g(x) \text{ has a double root at 1,2,3}$$

$$\therefore h(x) = a(x-1)^2(x-2)^2(x-3)^2$$

$$I = a \int_1^3 (x-1)^2(x-2)^2(x-3)^2 dx = \frac{32}{105}$$

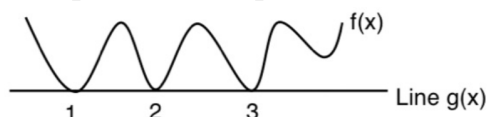
$$\text{Let } x-2 = t$$

$$I = a \int_{-1}^1 t^2(t+1)^2(t-1)^2 dt = \frac{32}{105}$$

$$I = a \int_{-1}^1 t^2(t^4 - 2t^2 + 1) dt = \frac{32}{105}$$

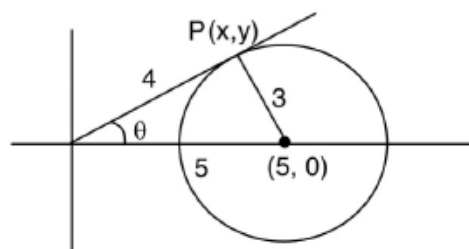
$$I = 2a \left[\frac{1}{7} - \frac{2}{5} + \frac{1}{3} \right] = \frac{32}{105}$$

$$= 2a \left[\frac{15 - 42 + 35}{105} \right] = \frac{32}{105}$$



$$a \times \frac{16}{105} = \frac{32}{105} \Rightarrow a = 2$$

3. Ans (3)



For largest value of $\frac{y}{x}$, line through origin

should touch circle in first quadrant

$$\Rightarrow \text{largest value of } \frac{y}{x}$$

$$= \text{largest value of } \tan \theta = \frac{3}{4}$$

4. Ans (1)

$$\text{Let } f(x) = x^{\frac{1}{x}}$$

$$\Rightarrow f'(x) = x^{\frac{1}{x}} \left(\frac{1 - \ln x}{x^2} \right)$$

If $x > e$ then $f(x)$ is decreasing

$$\therefore 2010 > 2009$$

$$\Rightarrow f(2010) < f(2009)$$

$$\Rightarrow (2010)^{\frac{1}{2010}} < (2009)^{\frac{1}{2009}}$$

$$\Rightarrow (2010)^{2009} < (2009)^{2010}$$

$$= \lim_{n \rightarrow \infty} \left\{ \left((2009)^{2010} \right)^n + \left((2010)^{2009} \right)^n \right\}^{\frac{1}{n}}$$

$$= \lim_{n \rightarrow \infty} (2009)^{2010} \left[1 + \left(\frac{(2010)^{2009}}{(2009)^{2010}} \right)^n \right]^{\frac{1}{n}} = (2009)^{2010}$$

5. Ans (2)

Given that,

$$f(x) = \begin{cases} -1 & , -2 \leq x \leq 0 \\ (x-1) & , 0 < x \leq 2 \end{cases}$$

Since, $x \in [-2, 2]$. Therefore, $|x| \in [0, 2]$

$$\Rightarrow f(|x|) = |x| - 1, \forall x \in [-2, 2]$$

$$\Rightarrow f(|x|) = \begin{cases} x-1 & , 0 \leq x \leq 2 \\ -x-1 & , -2 \leq x \leq 0 \end{cases}$$

$$\text{Also, } |f(x)| = \begin{cases} 1 & , -2 \leq x < 0 \\ 1-x & , 0 \leq x < 1 \\ x-1 & , 1 \leq x \leq 2 \end{cases}$$

Also, $g(x) = f(|x|) + |f(x)|$

$$g(x) = \begin{cases} -x & , -2 \leq x \leq 0 \\ 0 & , 0 \leq x < 1 \\ 2(x-1) & , 1 \leq x \leq 2 \end{cases}$$

$$\therefore g'(x) = \begin{cases} -1 & , -2 \leq x \leq 0 \\ 0 & , 0 \leq x < 1 \\ 2 & , 1 \leq x \leq 2 \end{cases}$$

$\Rightarrow g(x)$ is not differentiable at $x = 0$.

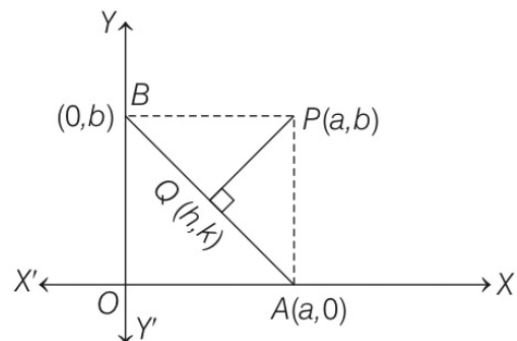
Hence, $g(x)$ is differentiable for all $x \in (-2, 2) - \{0, 1\}$

6. Ans (4)

Given, $AB = 8 = C$

$$\Rightarrow OA^2 + OB^2 = c^2 \Rightarrow a^2 + b^2 = c^2 \quad \dots(i)$$

Since, PQ is perpendicular to AB.



$\Rightarrow$ Slope of AB. Slope of PQ = -1

$$\Rightarrow \frac{0-b}{a-0} \cdot \frac{k-b}{h-a} = -1$$

$$\Rightarrow ah - bk = a^2 - b^2 \quad \dots(ii)$$

Equation of line AB is $\frac{x}{a} + \frac{y}{b} = 1$.

Since, Q lies on AB, therefore

$$\Rightarrow bh + ak = ab \quad \dots(iii)$$

On solving Eqs. (ii) and (iii), we get

$$\frac{h}{ab^2 + a(a^2 - b^2)} = \frac{k}{-b(a^2 - b^2) + a^2b} = \frac{1}{a^2 + b^2}$$

$$\Rightarrow \frac{h}{a^3} = \frac{k}{b^3} = \frac{1}{c^2} \quad [\text{from Eq. (i)}]$$

$$\Rightarrow a = (hc^2)^{1/3} \text{ and } b = (kc^2)^{1/3}$$

On substituting the values of a and b in

$$a^2 + b^2 = c^2,$$

$$\text{we get } h^{2/3} + k^{2/3} = c^{2/3}$$

Hence, locus of a point is $x^{2/3} + y^{2/3} = c^{2/3}$.

7. **Ans (2)**

Thus, equation of normal to $y^2 = 4x$ is
 $y = mx - 2m - m^3$, let it passes through (h, k) .

$$\text{or } m^3 + m(2 - h) + k = 0$$

Here, $m_1 m_2 m_3 = -k$ where $m_1 m_2 = \alpha$

$$\Rightarrow m_3 = -\frac{k}{\alpha}, \text{ it must satisfy Eq.(i)}$$

$$\Rightarrow -\frac{k^3}{\alpha^3} - \frac{k}{\alpha}(2 - h) + k = 0$$

$$\Rightarrow k^2 = \alpha^2 h - 2\alpha^2 + \alpha^3$$

$$\Rightarrow y^2 = \alpha^2 h - 2\alpha^2 + \alpha^3$$

On comparing with $y^2 = 4x$

$$\Rightarrow \alpha^2 = 4$$

$$\text{and } -2\alpha^2 + \alpha^3 = 0$$

$$\Rightarrow \alpha = 2$$

8. **Ans (3)**

$$\text{Total matrices} = {}^9C_5 \cdot {}^4C_2 = 756$$

$$|B| = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

$$= a_1 b_2 c_3 + a_2 b_3 c_1 + a_3 b_1 c_2 - a_1 b_3 c_2 - a_2 b_1 c_3 - a_3 b_2 c_1$$

$$|B| = 1$$

$$\Rightarrow 6 \text{ ways} \times (a_1 b_2 c_3 = 1 \text{ and all other five terms} = 0)$$

$$= 6 \times (\text{give 1 to non diagonal elements}) \times (a_1 b_2 c_3 = 1$$

$$\times -1 \times -1) = 6 \times 6 \times 3 = 108$$

similarly 108 ways for $|B| = -1$

$$|B| = 0$$

$\Rightarrow$ number of possible matrices

$$= 756 - 108 - 108 = 540$$

9. **Ans (1)**

$$f(x) = (\alpha x - 4)(x - 2) \Rightarrow f(x) = 0 \text{ has roots}$$

$$\frac{4}{\alpha}, 2$$

α must be negative and

$$\frac{4}{\alpha} \in [-4, -3] \Rightarrow \frac{\alpha}{4} \in \left(\frac{-1}{3}, \frac{-1}{4} \right] \Rightarrow \alpha \in \left(\frac{-4}{3}, -1 \right]$$

$$\Rightarrow a = \frac{-4}{3}, b = -1$$

$$\Rightarrow 9a^2 = 16, b^2 = 1 \Rightarrow 9a^2 + b^2 = 17$$

10. **Ans (8)**

$$B - A = (2\cot^{-1}(2) + 3\cot^{-1}(3)) - \left(\frac{1}{2}\cot^{-1}\left(\frac{1}{2}\right) + \frac{1}{3}\cot^{-1}\left(\frac{1}{3}\right) \right)$$

$$= 2(\cot^{-1}2 + \cot^{-1}3) + \cot^{-1}3 - \left(\frac{1}{3}\left(\cot^{-1}\frac{1}{2} + \cot^{-1}\frac{1}{3}\right) + \frac{1}{6}\cot^{-1}\frac{1}{2} \right)$$

$$= \frac{\pi}{2} + \cot^{-1}3 - \left[\frac{\pi}{4} + \frac{1}{6}\tan^{-1}2 \right] = \frac{\pi}{4} + \cot^{-1}3 - \frac{1}{6}\left[\frac{3\pi}{4} - \tan^{-1}3 \right]$$

$$= \frac{5\pi}{24} + \frac{5}{6}\cot^{-1}3$$

$$\text{Hence } a = 5; b = 24; c = 5; d = 6$$

$$a + b + c + d = 40$$

JEE(Main + Advanced) : LEADER & ENTHUSIAST COURSE - SCORE(ADVANCED)
ANSWER KEY
PAPER-2
PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	C	B	A	D	C	D
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,B,C	A,C	A,D	A,B,C	A,C	B,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	1.50 to .80	0.3 to .33	21.00	-90.62 or -90.63	10201.00	0.39

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	D	D	A	D	D	B
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,B,C,D	A,B,D	A,B,C	A,D	A,B,C,D	A
SECTION-II	Q.	1	2	3	4	5	6
	A.	1.50	4.81 or 4.82	7.00	1.25	7.00	2.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	B	B	D	D	B	C
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,B,C	A,B,D	A,B,C,D	A,B,C	A,B	B,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	11.00	0.00	1.00	2.01 or 2.02	0.50	5.00

HINT – SHEET
PART-1 : PHYSICS
SECTION-I (i)

1. Ans (C)

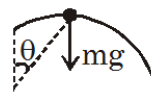


$$\phi_{\text{loop}} = 0 \text{ always}$$

$$\Rightarrow \varepsilon = 0$$

2. Ans (B)

Initial force on bead is outwards to balance mg .
By IIIrd law force on wire is inwards. Force is zero when



$$mg \cos \theta = \frac{mv^2}{R}$$

$$mgR - R \cos \theta = \frac{1}{2}mv^2$$

$$\Rightarrow \cos \theta = \frac{2}{3}$$

3. **Ans (A)**

$$f = \sigma E$$

$$f = \sigma \left(\frac{2R\lambda}{Kr} \right)$$

$$\frac{i}{A} = \frac{2\sigma R\lambda}{Kr}$$

$$\frac{-dq}{2\pi r} = \frac{\sigma q}{2\pi \epsilon_0 Kr}$$

$$-\int_{q_0}^q \frac{dq}{q} = \sigma \int_0^t \frac{dt}{\epsilon}$$

$$\ln \left(\frac{q}{q_0} \right) = -\frac{\sigma}{\epsilon} t$$

$$q = q_0 e^{-\left(\frac{\sigma}{\epsilon} t\right)}$$

4. **Ans (D)**

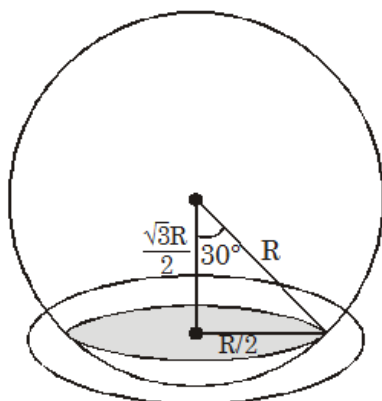
$$x_L = 0.5 R$$

$$\frac{1}{\omega C} = 0.5R$$

$$RC = \frac{1}{0.5\omega}$$

$$= \frac{2}{500} \times 1000$$

5. **Ans (C)**



$$q_{en} = \int_0^{R/2} \sigma_0 \left(1 - \frac{r}{R} \right) 2\pi r dr$$

$$= 2\pi\sigma_0 \int_0^{R/2} \left(r - \frac{r^2}{R} \right) dr$$

$$= 2\pi\sigma_0 \left[\frac{R^2}{8} - \frac{R^2}{24} \right]$$

$$= \frac{2\pi\sigma_0 R^2}{6}$$

6. **Ans (D)**

$$\frac{\lambda}{2} = 33.2$$

$$\Rightarrow \lambda = 66.4$$

$$\frac{3\lambda}{4} = \ell_2 + e$$

$$49.8 = 49 + e$$

$$e = 0.8 \text{ cm}$$

SECTION-I (ii)

7. **Ans (A,B,C)**

$$10 = \sqrt{\frac{k_1}{m}}, 20 = \sqrt{\frac{k_2}{m}}$$

$$\Rightarrow \frac{k_2}{k_1} = 4$$

$$\frac{b^2}{2m} = \frac{1}{2} \times k_1 a^2 = 5$$

$$\frac{R^2}{2m} = \frac{1}{2} k_2 R^2 = 10$$

$$4 \times \frac{R^2}{a^2} = 2$$

$$R = \frac{a}{\sqrt{2}}$$

$$\frac{b}{R} = \frac{1}{\sqrt{2}}$$

$$R = \sqrt{2}b = \frac{a}{\sqrt{2}}$$

$$2b = a$$

$$2ma \times 10 = a$$

$$m = 1 = 0.05 \text{ kg}$$

8. **Ans (A,C)**

$$\frac{Q}{4\pi\epsilon_0 r_0^2} = \frac{\sigma}{2\epsilon_0}$$

$$Q = 2\pi\sigma r_0^2 = 1\mu C$$

$$\frac{\sigma}{2\epsilon_0} = \frac{\lambda}{2\pi\epsilon_0 r_0}$$

$$\lambda = \pi\sigma = 0.5\mu C/m$$

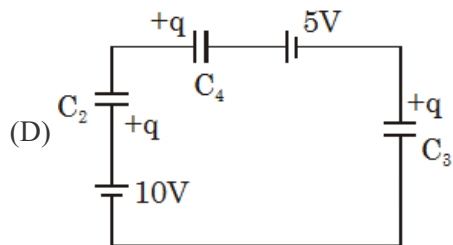
$$\frac{Q}{4\pi\epsilon_0 \times \left(\frac{1}{2}\right)^2} = k \frac{\lambda}{2\pi\epsilon_0 \left(\frac{1}{2}\right)}$$

$$Q = 1 = k \times 0.5 \Rightarrow k = 2$$

9. Ans (A,D)

$$(A) i = \frac{5}{160} = \frac{1}{32} A$$

(C) current is zero as S_2 is open



$$10 - \frac{q}{C_2} - \frac{q}{C_4} - 5 - \frac{q}{C_3} = 0$$

$$q \left(\frac{1}{C} + \frac{1}{C} + \frac{1}{C} \right) = 5$$

$$q = \frac{5C}{3}$$

10. Ans (A,B,C)

$$MR^2\omega = \left(MR^2 + \frac{M}{4}x^2 \right) \frac{8\omega}{9}$$

$$\frac{R^2}{8} = \frac{x^2}{4} \Rightarrow x = \frac{R}{\sqrt{2}}$$

$$\frac{1}{2}MR^2\omega^2 = \frac{1}{2} \times \frac{9}{8}MR^2 \times \left(\frac{8\omega^2}{9} \right) + \frac{1}{2} \frac{M}{8} \times v_x^2 \times 2$$

$$\frac{R^2\omega^2}{9} = \frac{v^2}{4}$$

$$\Rightarrow v = \frac{2R\omega}{3}$$

11. Ans (A,C)

$$\Delta p = \frac{h}{\lambda}$$

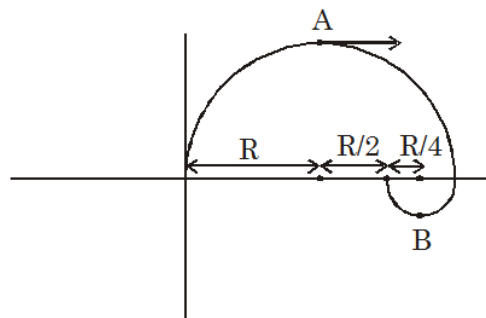
$$\frac{13.6 \left(\frac{1}{m^2} - \frac{1}{16} \right)}{13.6 \left(1 - \frac{1}{16} \right)} = \frac{7}{135}$$

$$\frac{1}{m^2} - \frac{1}{16} = \frac{15}{16} \times \frac{7}{135} = \frac{7}{16 \times 9}$$

$$m = 3$$

$$\frac{hC}{\lambda_e} = 13.6 \left(\frac{1}{9} - \frac{1}{16} \right)$$

12. Ans (B,C)



$$T = \frac{\pi m}{qB} + \frac{\pi m}{4qB} = \frac{5\pi m}{4qB}$$

$$(A) \frac{\pi m}{qB} = \frac{4T}{5}$$

(B) At $t = \frac{2T}{5}$ particle is at A

(C) At $t = \frac{9T}{10}$ particle is at B

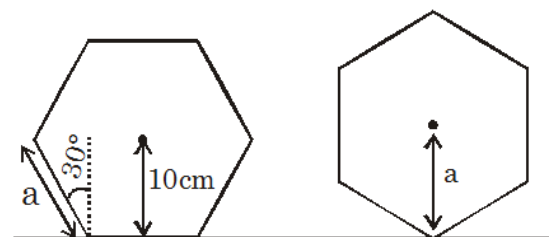
$$B \equiv \left(R + \frac{R}{2} + \frac{R}{4}, -\frac{R}{4} \right)$$

$$B \equiv \left(+\frac{7R}{4}, -\frac{R}{4} \right)$$

$$R = \frac{mv}{qB_1}, R = \frac{\pi mv}{qB_1}$$

SECTION-II

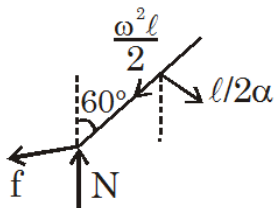
1. Ans (1.50 to 1.80)



$$a = \frac{10}{\cos 30^\circ} = \frac{10 \times 2}{\sqrt{3}}$$

$$\Delta h = \frac{10 \times 2}{\sqrt{3}} - 10$$

2. Ans (0.31 to 0.33)



$$mg \frac{l}{2} \sin 60 = \frac{m l^2}{3} \alpha$$

$$\alpha = \frac{3\sqrt{3}g}{4l}$$

$$mg \frac{l}{2} (1 - \cos 60) = \left(\frac{m l^2}{3} \right) \omega$$

$$\omega^2 = \frac{3g}{2l}$$

$$a_x = \frac{\omega^2 l}{2} \cos 30^\circ - \frac{l \alpha}{2} \cos 60^\circ$$

$$\frac{3\sqrt{3}g}{8} - \frac{3\sqrt{3}g}{16} = \frac{3\sqrt{3}}{16}$$

$$f = m a_x = \frac{3\sqrt{3}}{16} mg$$

3. Ans (21.00)

$$T \cos \theta = mg$$

$$T \sin \theta = \frac{q^2}{4\pi\epsilon_0 r^2}$$

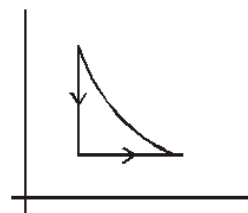
$$\tan \theta = \frac{q^2}{4\pi\epsilon_0 mgr^2}$$

$$mg = \epsilon_r (mg - B)$$

$$\epsilon_r = \frac{\rho_b}{\rho_b - \rho_L} = \frac{840}{40} = 21$$

4. Ans (-90.62 or -90.63)

$$\gamma = \frac{5}{3}$$



$$\Delta Q_1 = n C_V \Delta T = \frac{3}{2} [P_f - P_i] V_i$$

$$\Delta Q_2 = n C_P \Delta T = \frac{5}{2} P_f [V_f - V_i]$$

$$\Delta Q = \frac{3}{2} \times 10^{-3} \times \left(\frac{1}{3^2} - 1 \right) \times 10^5$$

$$+ \frac{5}{2} \times \frac{1}{32} \times 10^5 (8 - 1) \times 10^3$$

$$= -\frac{9300}{64} + \frac{3500}{64} = -90.625$$

5. Ans (10201.00)

$$\frac{4\pi R^3}{3} = k \times \frac{4}{3} \pi r^3$$

$$r = \frac{R}{k^{1/3}}$$

$$10^{-3} = \frac{S \times 4\pi r^2}{k^{2/3}} \times k - S \times 4\pi R^2$$

$$10^{-3} = S \times 4\pi R^2 [k^{1/3} - 1]$$

$$= \frac{0.1}{4\pi} \times 4\pi \times 10^{-4} [k^{1/3} - 1]$$

$$k^{1/3} = 101$$

$$r = \frac{R}{101}$$

$$v \propto r^2$$

6. Ans (0.39)

$$\frac{3kA}{1} (T - 10) = \frac{kA}{1} (400 - T)$$

$$5T - 50 = 400 - T$$

$$6T = 450$$

$$\tau = 75$$

$$T = 10 + 65 x$$

$$d\ell = 1.2 \times 10^{-5} \times 65 \int_0^1 x' dx$$

$$= 0.6 \times 10^{-5} \times 65$$

$$= 39 \times 10^{-5} \text{ m}$$

PART-2 : CHEMISTRY

SECTION-I (i)

3. Ans (A)

Assuming steady, state ;

$$\frac{d[B]}{dt} = k_1[A] - k_2[B] = 0 \Rightarrow k_1[A] = k_2[B] \dots(1)$$

$$\frac{d[C]}{dt} = k_2[B] - k_3[C] = 0 \Rightarrow k_2[B] = k_3[C] \dots(2)$$

$$\frac{d[D]}{dt} = k_3[C] - k_4[D] = 0 \Rightarrow k_3[C] = k_4[D] \dots(3)$$

from (1), (2) & (3)

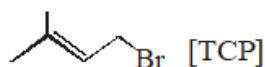
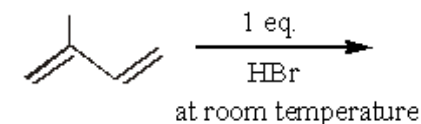
$$k_1[A] = k_4[D] \Rightarrow [D] = \frac{k_1}{k_4} [A]_0 e^{-K_1 t}$$

$$= \frac{0.1386}{6.93 \times 10^{10}} \times 1 \times e^{-1.386}$$

4. Ans (D)

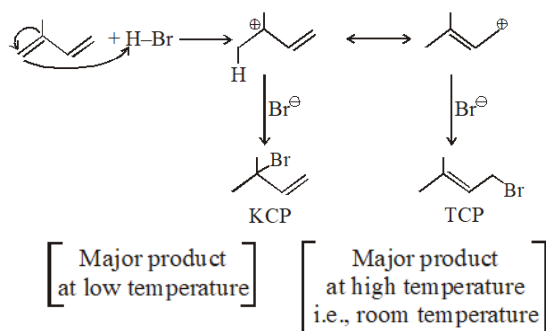
Refer theory

5. Ans (D)

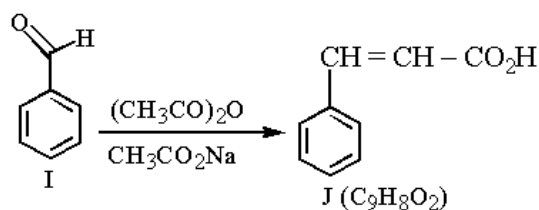


Major product

Mechanism :



6. Ans (B)



J gives effervescence on treatment with NaHCO₃ due to presence of -CO₂H group and positive Baeyer's test due to presence of unsaturation.

SECTION-I (ii)

7. Ans (A,B,C,D)

$$(a) \frac{P^0 - P}{P} = \frac{n_1}{n_2} \Rightarrow \frac{760 - 740}{740} = \frac{1}{n_2} \Rightarrow n_2 = 37$$

∴ Moles of water separated as ice = 200 - 37 = 163

$$(b) \Delta T_f = K_f \cdot m = 2.0 \times \frac{1}{(37 \times 18)/1000} = \frac{2000}{37 \times 18} K$$

$$\therefore T = \left(273 - \frac{2000}{37 \times 18} \right) K$$

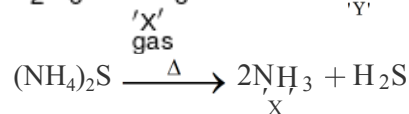
(c) For original solution

$$\Delta T_f = K_f \cdot m = 2 \times \frac{1}{(37 \times 18)/1000}$$

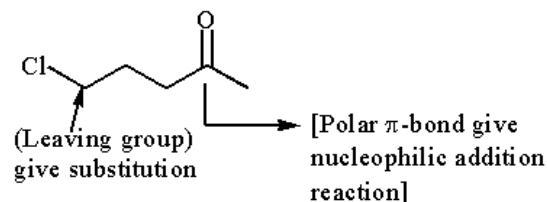
$$\therefore F.P = 0 - \Delta T_f = \frac{10}{18} C$$

$$(d) \text{ For final solution : } \frac{P^0 - P}{P^0} = X_1 = \frac{1}{1 + 37} = \frac{1}{38}$$

9. Ans (A,B,C)

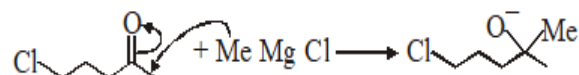


11. Ans (A,B,C,D)

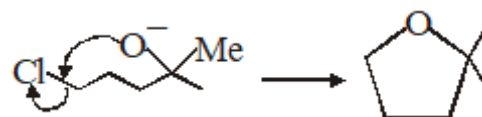


(i) Grignard prefer to give nucleophilic addition

on polar π-bond and form anion intermediate.

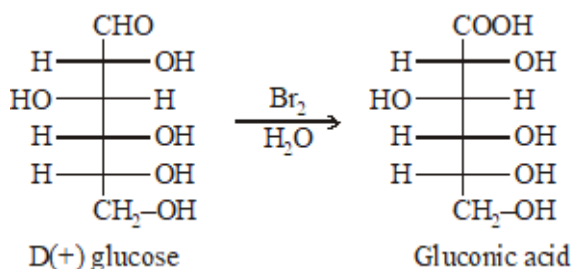


(ii) In next step anion give intramolecular nucleophilic substitution reaction & form 5 membered ring.



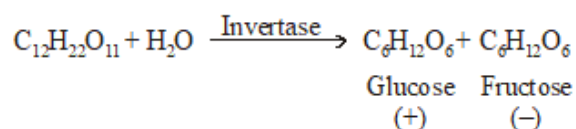
12. Ans (A)

(A) FALSE :



(B) TRUE : Six member hemiacetal on anomeric carbon gives α-D glucose & β-D glucose.

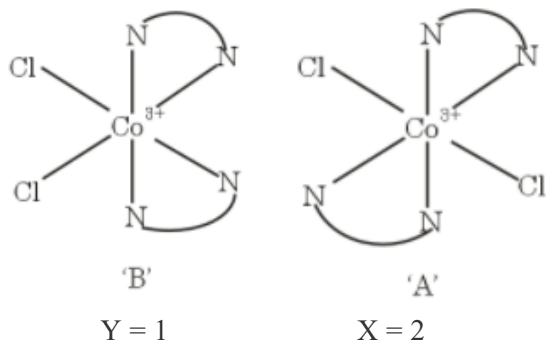
(C) TRUE :



(D) TRUE : Monosaccharide cannot be hydrolysed to give polyhydroxy aldehydes and ketones

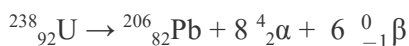
SECTION-II

1. Ans (1.50)



$$\frac{X + Y}{2} = \frac{2 + 1}{2} = 1.50$$

2. Ans (4.81 or 4.82)



$$t = 0 \quad 1.2 \text{ mol}$$

$$t = \infty \quad 0 \quad 1.2 \text{ mol} \quad 8 \times 1.2 \text{ mol}$$

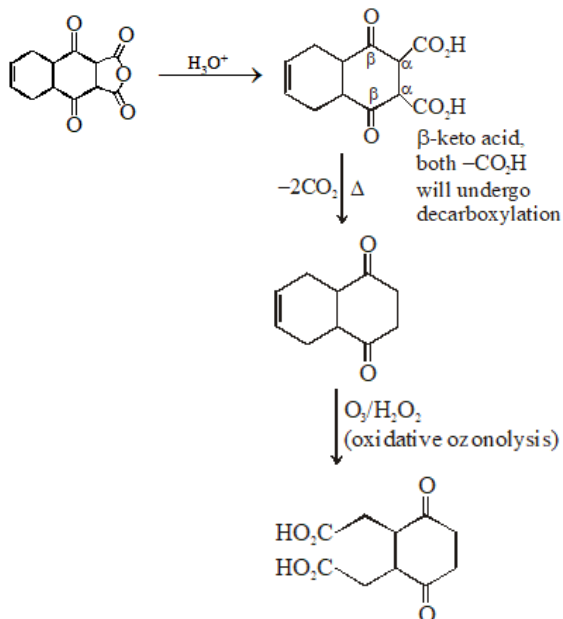
$$\Rightarrow \frac{P_t}{P_i} = \frac{n_f}{n_i} = \frac{(8 \times 1.2 + 1)}{2.2} = 4.82$$

4. Ans (1.25)

$$\text{Incident energy} = \left(\frac{1240}{300} \right) \text{ eV} = 4.13$$

$$\therefore x = 4 ; y = 5 \Rightarrow \frac{y}{x} = 1.25$$

6. Ans (2.00)



Number of carboxylic acids groups present in final product = 2.

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (B)

$$P = I + B, \quad B = \begin{bmatrix} 0 & 0 & 0 \\ 4 & 0 & 0 \\ 8 & 4 & 0 \end{bmatrix}$$

$$B^2 = \begin{bmatrix} 0 & 0 & 0 \\ 4 & 0 & 0 \\ 8 & 4 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 4 & 0 & 0 \\ 8 & 4 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 16 & 0 & 0 \end{bmatrix}$$

$$B^3 = 0$$

$$P^{500} = (I + B)^{500} = I + 500B + 250 \times 499B^2$$

$$\Rightarrow P^{500} - I = 500B + 250 \times 499B^2 \frac{q_{31} + q_{32}}{q_{21}}$$

$$= \frac{500 \times 8 + 250 \times 499 \times 16 + 500 \times 4}{500 \times 4 + 0} = 1001$$

2. Ans (B)

$$\frac{x-1}{1} = \frac{y-2}{-1} = \frac{z-3}{1} = \frac{-2 \times 3}{3}$$

$$\Rightarrow P(-1, 4, 1), Q(3, 1, 7)$$

$$\vec{N} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & -3 & 6 \\ 1 & 2 & 1 \end{vmatrix} = -15\hat{i} + 2\hat{j} + 11\hat{k}$$

$$\text{Required plane is : } 15x - 2y - 11z + 34 = 0$$

3. Ans (D)

For option (A),

$$e^x - \int_0^x f(t) \sin t \, dt > 0, \forall x \in (0, 1)$$

For option (B)

$$1 + x^{2021} - (f(x))^{2022} > 0, \forall x \in (0, 1)$$

For option (C), let

$$f(x) + \int_0^{\frac{\pi}{2}} f(t) \cdot \sin t \, dt > 0, \forall x \in (0, 1)$$

For option (D),

$$g(x) = 1 - x^{2021} - (f(x))^{2022}$$

$$g(1) < 0$$

$$g(0) > 0$$

$$\Rightarrow g(x) = 0 \text{ has at least one solution in } (0, 1)$$

4. **Ans (D)**

$$\Delta LMN = 16 \Rightarrow \frac{1}{2} \times 2b \times a = 16$$

$$\Rightarrow ab = 16$$

$$a = b \Rightarrow a = b = 4$$

Area of triangle formed by the tangent and asymptotes of H = 16

$$e = \sqrt{2}$$

product of lengths of perpendicular drawn from $(-4\sqrt{2}, 0)$ and $(4\sqrt{2}, 0)$ upon any tangent = 16

5. **Ans (B)**

We have $\text{trace}(AB) = \text{trace}(BA)$

$$\Rightarrow \text{Trace}(X) = \sum_{K=1}^5 \text{Trace}(P_K P P_K^T)$$

$$= \sum_{K=1}^5 \text{Trace}(P_K P_K^T P)$$

$$= \sum_{K=1}^5 \text{Trace}(P) = 90$$

As $\text{Trace}(X) \neq 0$

$\Rightarrow X$ can not be skew-symmetric matrix.

6. **Ans (C)**

We have for $k < x < k+1$

$$\frac{1}{x+1} < \frac{k+1}{x(x+1)} < \frac{1}{x}$$

$$\Rightarrow \int_k^{k+1} \frac{dx}{x+1} < \int_k^{k+1} \frac{(k+1)dx}{x(x+1)} < \int_k^{k+1} \frac{dx}{x}$$

$$\Rightarrow \ln(k+2) - \ln(k+1) < I_k < \ln(k+1) - \ln k$$

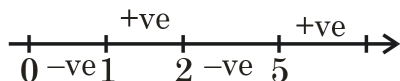
$$\Rightarrow \ln 1000 - \ln 2 < I < \ln 999 - \ln 1$$

$$\Rightarrow \ln 500 < I < \ln 999$$

SECTION-I (ii)

7. **Ans (A,B,C)**

Sign scheme for $F'(x)$



$$F(x) = \int_0^x (x^5 - 5x^4 - 5x^3 + 25x^2 - 4x - 20) dx$$

$$= \frac{x^6}{6} - x^5 - \frac{5x^4}{4} + \frac{25x^3}{3} - 2x^2 - 20x$$

$$F(2) = \frac{232}{3} - 100 < 0$$

8. **Ans (A,B,D)**

$R \equiv$ red ball is chosen

$$P(R) = \sum_{i=1}^4 P(B_i)P\left(\frac{R}{B_i}\right)$$

$$= \frac{2}{10} \times \frac{5}{10} + \frac{2}{10} \times \frac{3}{10} + \frac{3}{10} \times \frac{7}{10} + \frac{3}{10} \times \frac{4}{10}$$

$$= \frac{10 + 6 + 21 + 12}{100} = \frac{49}{100}$$

$$P(B_4 \cap R) = \frac{3}{10} \times \frac{4}{10}$$

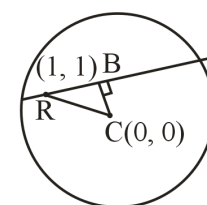
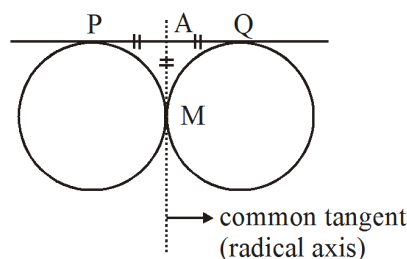
$$= \frac{12}{100} = \frac{3}{25}$$

$$P\left(\frac{R}{B_4}\right) = \frac{4}{10} = \frac{2}{5}$$

$$P\left(\frac{B_4}{R}\right) = \frac{3/25}{49/100} = \frac{12}{49}$$

9.

Ans (A,B,C,D)



$$AP = AQ = AM$$

Locus of M is a circle having PQ as its

diameter

$$E_1 : x^2 + y^2 = 4, x \neq \pm 2$$

E_2 : Locus of mid point of chord passing

through (1, 1) is $x^2 + y^2 - x - y = 0$ (0, 2), (2, 0)

and (1, 1) are collinear

$\Rightarrow$ option (D) is correct.

10. Ans (A,B,C)

Condition for tangency

$$3^2 = a^2 - 16$$

$$\Rightarrow |a| = 5$$

11. Ans (A,B)

Apply LMVT on $(-8, 0)$

$$\frac{f(0) - f(-8)}{0 - (-8)} = f'(x_0)$$

for some x_0 in $(-8, 0)$

$$\Rightarrow |f'(x_0)| \leq \frac{16}{8} = 2$$

Suppose $G(x) = (f(x))^2 + (f'(x))^2, \forall x \in \mathbb{R}$

$$G(0) = 2021$$

For some $x_1 \in (-8, 0)$ and $x_2 \in (0, 8)$

$$G(x_1) \leq 68, G(x_2) \leq 68$$

For $x \in (x_1, x_2)$, maximum value of $G(x) \geq 2021$

$$\Rightarrow \text{maximum of } G(x) \geq 2021 \forall x \in (x_1, x_2)$$

Suppose maximum of $G(x)$ occurs at $x = \alpha$,

$$\alpha \in (x_1, x_2)$$

$$\Rightarrow G'(\alpha) = 0$$

$$\Rightarrow 2f(\alpha)f'(\alpha) + 2f'(\alpha)f''(\alpha) = 0$$

$$\text{At } x = \alpha, f(\alpha) \neq 0$$

$$\text{as } (f(\alpha))^2 + (f'(\alpha))^2 \geq 2021$$

$$\Rightarrow f(\alpha) + f'(\alpha) = 0$$

As $f'(\alpha) \neq 0$ and $f(x)$ is twice differentiable, there

will exist an interval on which $f(x)$ is one-one.

12. Ans (B,D)

$$f(x) = 1 - 2x + \int_0^x e^{x-t} f(t) dt$$

$$\Rightarrow e^{-x} f(x) = e^{-x} (1 - 2x) + \int_0^x e^{-t} f(t) dt$$

Differentiate w.r.t. x .

$$-e^{-x} f(x) + e^{-x} f'(x) = -e^{-x} (1 - 2x) + e^{-x} (-2) + e^{-x} f(x)$$

$$\Rightarrow -f(x) + f'(x) = -(1 - 2x) - 2 + f(x).$$

$$\Rightarrow f'(x) - 2f(x) = 2x - 3$$

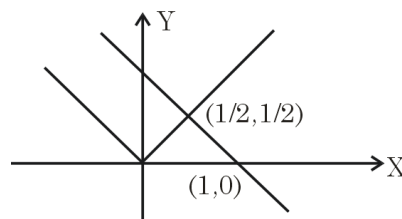
Integrating factor = e^{-2x} .

$$\begin{aligned} f(x) \cdot e^{-2x} &= \int e^{-2x} (2x - 3) dx \\ &= (2x - 3) \int e^{-2x} dx - \int \left((2) \int e^{-2x} dx \right) dx \\ &= \frac{(2x - 3)e^{-2x}}{-2} - \frac{e^{-2x}}{2} + c \end{aligned}$$

$$f(x) = \frac{2x - 3}{-2} - \frac{1}{2} + ce^{2x}$$

$$f(0) = \frac{3}{2} - \frac{1}{2} + c = 1 \Rightarrow c = 0$$

$$\therefore f(x) = 1 - x$$



$$\text{Area} = \frac{1}{2} \times 1 \times \frac{1}{2} = \frac{1}{4}$$

SECTION-II

1. Ans (11.00)

$$\lim_{x \rightarrow 0} \frac{\alpha x + \beta x^3 - x + \frac{x^3}{3!} - \frac{x^5}{5!} \dots}{x^3} = 1$$

$$\Rightarrow \alpha = 1, \beta + \frac{1}{3!} = 1$$

$$\Rightarrow \alpha = 1, \beta = 1 - \frac{1}{6} = \frac{5}{6}$$

$$\text{So, } 6(\alpha + \beta) = 6 + 5 = 11$$

2. **Ans (0.00)**

$$\vec{c} \cdot \vec{a} = x\vec{a} \cdot \vec{a} + y\vec{b} \cdot \vec{a} + (\vec{a} \times \vec{b}) \cdot \vec{a}$$

$$\Rightarrow x = 2\cos\alpha$$

Similarly, $y = 2\cos\alpha$

$$|\vec{c}|^2 = (4\cos^2\alpha) \times 4 + (4\cos^2\alpha) \times 4 + 16$$

$$\Rightarrow \cos^2\alpha = 0$$

3. **Ans (1.00)**

$$f(-x) = -f(x)$$

$$I = \int_{-\pi/3}^{\pi/3} \frac{dx}{(1 + (2021)^{f(x)})(2 - \cos 2x)}$$

Apply King's Rule

$$I = \int_{-\pi/3}^{\pi/3} \frac{(2021)^{f(x)} dx}{((2021)^{f(x)} + 1)(2 - \cos 2x)}$$

$$\Rightarrow I + I = \int_{-\pi/3}^{\pi/3} \frac{dx}{2 - \cos 2x}$$

$$\Rightarrow I = \int_{-\pi/3}^{\pi/3} \frac{dx}{2 - \frac{1 - \tan^2 x}{1 + \tan^2 x}} = \int_0^{\pi/3} \frac{\sec^2 x dx}{1 + 3\tan^2 x}$$

$$= \frac{1}{\sqrt{3}} (\tan^{-1} 3)$$

$$\Rightarrow [\sqrt{3} I] = [\tan^{-1} 3] = 1$$

4. **Ans (2.01 or 2.02)**

We equate the general terms of three respective

$$\text{A.P.'s as } 1 + 3a = 2 + 5b = 3 + 7c$$

$$\Rightarrow 3 \text{ divides } 1 + 2b \text{ and } 5 \text{ divides } 1 + 2c$$

$$\Rightarrow 1 + 2c = 5, 15, 25 \text{ etc.}$$

$$\text{So, first such terms are possible when } 1 + 2c = 15 \text{ i.e. } c = 7$$

$$\text{Hence, first term} = a = 52$$

$$d = \text{lcm}(3, 5, 7) = 105$$

5. **Ans (0.50)**

$$A(1, 0, 0), B\left(\frac{1}{2}, \frac{1}{2}, 0\right) \text{ \& } C\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$$

$$\vec{OA} = \hat{i}, \vec{OB} = \frac{1}{2}\hat{i} + \frac{1}{2}\hat{j},$$

$$\vec{OC} = \frac{1}{3}\hat{i} + \frac{1}{3}\hat{j} + \frac{1}{3}\hat{k}$$

$$\Rightarrow V = \frac{1}{36}$$

6. **Ans (5.00)**

$$\sqrt{3} \cos x + \frac{2b}{a} \sin x = \frac{c}{a}$$

$$\text{Now, } \sqrt{3} \cos \alpha + \frac{2b}{a} \sin \alpha = \frac{c}{a} \quad \dots (1)$$

$$\sqrt{3} \cos \beta + \frac{2b}{a} \sin \beta = \frac{c}{a} \quad \dots (2)$$

$$\sqrt{3} [\cos \alpha - \cos \beta] + \frac{2b}{a} (\sin \alpha - \sin \beta) = 0$$

$$\sqrt{3} \left[-2 \sin \left(\frac{\alpha + \beta}{2} \right) \sin \left(\frac{\alpha - \beta}{2} \right) \right] + \frac{2b}{a} \left[2 \cos \left(\frac{\alpha + \beta}{2} \right) \sin \left(\frac{\alpha - \beta}{2} \right) \right] = 0$$

$$-\sqrt{3} + 2\sqrt{3} \cdot \frac{b}{a} = 0$$

$$\Rightarrow a = 2b \Rightarrow a^2 + b^2 = 5b^2 \geq 5$$